Fiscal Year:	FY 2010	**Task Last Updated:**	FY 11/02/2012
PI Name:	Gernhardt, Michael Ph.D.		
Project Title:	Comparison of Biodynamics Models Using Automotive Racing Crash Data		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	michael.l.gernhardt@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	281-244-0125
Organization Name:	NASA Johnson Space Center		
PI Address 1:	2101 NASA Parkway, Code ER		
PI Address 2:	EVA Physiology Laboratory / Advanced Exploration Systems		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	01/01/2008	**End Date:**	09/30/2010
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Given the level of fidelity of the National Association of Stock Car Auto Racing (NASCAR) data and thorough documentation of injuries, there should be a strong correlation between driver injury and biodynamic response. Additionally, since NASCAR shares more similarities with Orion (NASA's Multi-Purpose Crew Vehicle) than do passenger car occupant protection systems, the probability of injury determined from NASCAR would be better correlated to actual injury risk for Orion.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:			

Task Progress:	Via a Space Act Agreement, the Occupant Protection (OP) team obtained data on race car crash event dynamics and associated injuries for all recorded impacts during the 2002 to 2008 racing seasons, including impacts during race events, qualifying heats, and practice. The data included detailed information about resulting injuries (if any) and associated abbreviated injury scale (AIS) ratings, including any intrusion into the occupant compartment, which could affect the relevance and magnitude of recorded injuries and accelerations. NASCAR provided 4015 non-attributable impact files, including 44 with injury information. Vehicle dynamics were recorded on each vehicle when acceleration exceeded 4.5 G. The processed data were analyzed to calculate peak acceleration, change in velocity, pulse length, and other appropriate metrics. Dynamic Finite Element (FE) analysis software LS-DYNA (Livermore Software Technology Corporation 2009) was used with a 50th percentile male Hybrid III Anthropomorphic Test Device (ATD) finite element model (FEM) developed and validated by Humanetics Innovative Solutions, Inc. The FEM configured a NASCAR seat (generalized representation) with head and neck restraint device, 5-point harness, helmet, and head surround.
Bibliography Type:	Description: (Last Updated: 10/31/2019)
Articles in Peer-reviewed Journals	Somers JT, Granderson B, Melvin JW, Tabiei A, Lawrence C, Feiveson A, Gernhardt M, Ploutz-Snyder R, Patalak J. "Development of head injury assessment reference values based on NASA injury modeling." Stapp Car Crash J. 2011 Nov;55:49-74. PMID: 22869304 , Nov-2011